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The Greenness-of-Jobs Index (GOJI): A Task-Based Approach to Measure the Green Transition Exposure and the Greening of Occupations

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Abstract

Global efforts toward a low-carbon and greener economy are reshaping work by altering the task content of occupations. This paper introduces the *Greenness-of-Jobs Index (GOJI)*, a task-based indicator that quantifies the share of environment-related tasks in each occupation in Germany. Using task information from the BERUFENET database, I apply natural language processing to classify tasks as green, brown, or white based on international standards and to calculate their occupational-level shares. I trace GOJI’s evolution—from its initial version in 2018 (GOJI 1.0) to the current GOJI 2.0 series, available annually from 2012 to 2024. Using concrete examples, I demonstrate the classification process and coding logic. I situate GOJI within the literature on green jobs, skills, and tasks, and highlight its methodological advances over output-based and previous task-based indices. GOJI’s task-level design supports strategic foresight into the technological and social transitions underpinning sustainable labor market transformation.

Keywords: Green jobs; Task-based index; Occupational tasks; Environmental sustainability; Labor market transition; Natural language processing

1 Introduction

Climate change and environmental degradation have triggered widespread efforts toward greener economies and sustainable development. As industries implement cleaner technologies and more resource-efficient processes, the nature of work is changing. Jobs across diverse sectors are evolving to include “green” tasks oriented toward environmental protection, energy efficiency, and pollution reduction—or to leverage the potential of existing green tasks. International institutions such as the [International Labour Organization](#) (2018) and the [Organisation for Economic Co-operation and Development](#) (2023) emphasize that these developments have profound implications for the world of work. The need to measure the environmental content of jobs is therefore pressing, as highlighted by frameworks developed by the [United Nations Environment Programme](#) (2011). These transformations raise key questions for labor markets: To what extent are occupations becoming greener, which remain entrenched in environmentally harmful activities, and what does this mean for employment, wages, and the demands for vocational and continuing education? Addressing these questions requires robust, scalable, and policy-relevant measurement approaches. Reliable metrics are essential to anticipate workforce needs, monitor transition pathways, and guide vocational and innovation policy.

Defining and measuring “green jobs” has proven challenging. Early efforts often focused on sectors or outputs, labeling entire jobs or industries as green based on the production of environmental goods and services. For example, the U.S. Bureau of Labor Statistics (BLS) distinguished between jobs producing green outputs (e.g., renewable energy equipment) and jobs involving green processes (e.g., roles that improve a firm’s environmental performance) ([U.S. Bureau of Labor Statistics, 2013](#)). International bodies like the International Labour Organization (ILO) offered broader definitions, adding that truly *green jobs* should also meet standards of decent work ([International Labour Organization, 2018](#)). However, it has proven difficult to measure both aspects together, and this approach may have impeded efforts to quantify green jobs. All output-based approaches struggle to capture the *greening of occupations* within traditional sectors. Many conventional jobs are gradually incorporating environmentally friendly practices, even if

their primary output is not explicitly green but rather environmentally neutral or even “brown.” Process-oriented approaches were introduced to account for this integrated environmental activity([Consoli et al., 2016](#)). Yet, existing metrics still often treat jobs as monolithically green or not, without capturing gradations of environmental content.

This paper advances a task-based approach to measuring the greenness of jobs. Rather than classifying entire occupations or industries as green, I focus on the composition of tasks within each occupation. Tasks are the unit of work activities that produce outputs, and they can change over time even if an occupation’s title remains the same. I differentiate tasks that are explicitly environmental (“green tasks”) from those that are environmentally detrimental (“brown tasks”) or neutral (“white tasks”). By quantifying the share of green and brown tasks in each occupation, I construct a continuous **Greenness-of-Jobs Index (GOJI)** for every occupation and track how it evolves. This task-based index provides a more granular and dynamic measure of occupational greenness than prior binary labels. Analogous to the skills taxonomies used by the European Commission and the OECD, the GOJI differentiates between “green tasks,” which are potentially beneficial to the environment or climate, “white tasks,” which are considered environmentally neutral, and “brown tasks,” which are potentially harmful to environmental or climate goals ([European Commission, 2022](#), [Organisation for Economic Co-operation and Development, 2023](#)). This typology provides a conceptual foundation for assessing occupational roles in the green transition of labor markets.

The first version of the GOJI was originally published in 2018/2019 focusing only on green tasks and working solely with text mining results for a period of time from 2012 to 2016 ([Janser, 2018, 2019](#)). It has since undergone significant refinements. In this report, I present an expanded methodology for GOJI, termed **GOJI v2**. I begin by laying out the conceptual foundations of green, brown, and white tasks, including how my definitions align with and diverge from international approaches to defining green jobs and related concepts. Next, I recount the methodological development of GOJI: from the initial GOJI v1 from 2018 to a beta version with an improved classification of brown tasks and expanded data until 2022, and finally to the current GOJI v2 approach with data until

2024. I then detail my coding methodology, which uses natural language processing to identify environmental-friendly and environmental-harmful content in occupational tasks lists, illustrating the logic with examples drawn from actual occupations.

To contextualize my contribution, I provide a comprehensive literature review of previous research on jobs, occupations, tasks and skills in the green transition. I survey studies in economics, sociology and related fields that have sought to measure green and brown employment, examine skill needs in green versus non-green jobs, and analyze labor market impacts of the green transition. This review highlights how GOJI builds on earlier work and addresses gaps by offering a dynamic, task-level measure of occupational greenness.

My findings indicate that the greenness-of-jobs can be measured with considerable detail, revealing a wide spectrum from “dark green” occupations rich in eco-friendly tasks to “dark brown” occupations dominated by polluting tasks. I observe that many occupations lie in between, having a mix of white tasks with some green or brown elements. Tracking these changes over time (the *greening of occupations*) can yield new insights into how the workforce adapts to environmental challenges. Such insights have policy relevance: they can inform education and training (by identifying skills needed as occupations green), and help target transition assistance to workers in brown jobs.

The remainder of this report is organized as follows. Section 2 defines green, brown, and white tasks and situates my approach in the context of definitional debates and international measurement efforts. Section 3 describes the evolution of the GOJI methodology from its inception to the present, highlighting key improvements and rationales. Section 4 provides a detailed account of the data and coding methods used in GOJI 2025, including the text-mining algorithm and examples of task classification. Section 5 offers a review of relevant literature on green jobs and related concepts as well as on labor market impacts of environmental transition. Section 6 concludes with a discussion of implications and areas for further research. Extensive appendices follow, which include classification tables, task coding instructions, and a side-by-side comparison of different GOJI versions.

2 Conceptual Foundations: Green, Brown, and White Tasks

2.1 Defining Green, Brown and White Tasks

A central premise of my approach is that jobs can be understood as bundles of tasks, some of which have an environmental orientation and others do not. I define **green tasks** as those work activities that directly contribute to environmental protection, resource conservation, or pollution reduction in the context of producing goods or services. Conversely, **brown tasks** are work activities that involve practices harmful to the environment, such as the use of fossil fuels or energy-intensive, polluting processes. Tasks that are neither explicitly green nor brown—those that have no clear environmental impact—are considered **white tasks** (sometimes referred to as “white tasks”).

These definitions align with the broader notion of distinguishing environmentally beneficial versus harmful aspects of work. Formally, I adopt and slightly adapt the definitions put forth in Janser (2019). An *environment-friendly (green) task* is any occupational requirement—core duty or ancillary task—that is related to improving environmental outcomes. This can include tasks anywhere along a firm’s value chain, from product design and production to logistics and waste management, so long as the task’s content aims to reduce the use of fossil fuels, decrease greenhouse gas emissions, increase energy or material efficiency, or support the adoption of renewable energy or environmental technologies. Examples of green tasks include *developing strategies to improve energy efficiency in manufacturing processes, installing and maintaining solar panels or other renewable energy systems, monitoring air or water quality to ensure environmental compliance, and implementing recycling and waste reduction programs* within an organization.

In contrast, an *environment-harmful (brown) task* is defined as an occupational requirement that involves practices with negative environmental impacts. Brown tasks typically relate to the production or use of carbon-intensive energy and materials, the emission of pollutants, or processes that impede environmental protection efforts. They include activities centered on fossil fuel extraction and usage (for instance, *operating coal-*

fired power plant boilers or *drilling for oil and gas*), tasks in which greenhouse gas emissions are a direct output (such as *managing blast furnace operations in steel production*), and other practices that inherently cause pollution or high resource consumption. For example, *applying chemical pesticides in agriculture* can be considered a brown task due to its environmental harm, as can *operating diesel-powered heavy machinery* in construction or logistics. These tasks contribute to environmental degradation either by emitting pollutants or by sustaining dependence on non-renewable, high-emission resources.

Many tasks in the economy are **neutral**, meaning they do not have an obvious positive or negative environmental dimension. These might be general administrative, managerial, or technical tasks that could be present in any job regardless of industry (e.g. *managing a team*, *preparing financial accounts*, or *training new staff*). White tasks neither advance environmental goals nor significantly worsen environmental outcomes. In my framework, white tasks form the baseline of an occupation’s task set, against which the presence of green or brown tasks is measured.

By classifying tasks into these three categories, I acknowledge that greenness is not an all-or-nothing property of a job, but a matter of degree. An occupation might have a mix of task types. For instance, consider a building electrician: if they spend part of their time installing energy-efficient LED lighting or solar panels (green tasks) and part of their time wiring conventional electrical systems (white tasks), that occupation has a certain green task share. On the other hand, a coal plant maintenance technician might perform equipment repair (neutral) but also handle coal combustion systems (brown tasks), giving that occupation a brown task component. The concept of “greening of jobs” refers precisely to changes in this mix—if the electrician’s role evolves to focus more on renewable energy systems over time, its green share rises; if the coal technician’s role shifts towards managing emissions control equipment (slightly greener tasks) or if the occupation itself declines, the labor market is said to green.

It is worth noting that my use of the terms *green* and *brown* tasks parallels how some studies describe “clean” versus “dirty” industries or jobs. However, I apply it at the granular task level. This is conceptually powerful because it allows virtually

any occupation to have a greenness dimension—unlike earlier definitions of *green jobs* that might categorically exclude, say, accountants or drivers, unless those jobs are in a renewable energy company. Under a task-based approach, even an accountant’s job could have a green component (if, for example, they specialize in carbon accounting or sustainability reporting, those tasks would be green) or a brown component (though in an office job this is less likely unless indirectly via the sector’s nature). In most cases, occupations like accountants would mainly have white tasks.

2.2 Key Concepts: Green Transition Exposure and Greening of Occupations

Green transition exposure refers to the extent to which an occupation is affected—directly or indirectly—by the societal shift toward an environmentally sustainable economy. This exposure may stem from regulatory changes, the adoption of green technologies, the phasing out of carbon-intensive activities, or shifts in consumer preferences and production processes. Importantly, exposure can be positive (e.g., increased demand for renewable energy technicians) or negative (e.g., declining relevance of coal-related tasks). In the context of GOJI, green transition exposure is captured by the presence and weight of *green* and *brown* tasks in an occupation’s profile, indicating both its potential for transformation and the direction of that transformation.

The *greening of an occupation* denotes a structural change in its task composition over time, whereby environmentally beneficial (“green”) tasks become more prevalent relative to neutral or environmentally harmful (“brown”) tasks. This concept highlights the dynamic nature of occupational content: even jobs not traditionally classified as “green” may increasingly include sustainability-related responsibilities, such as energy efficiency, waste reduction, or sustainable resource use. The GOJI framework measures greening as a shift in the task-level index for a given occupation across different time points, thus quantifying how jobs evolve in response to ecological and technological change.

2.3 International Approaches and Definitional Debates

The delineation of green vs. brown tasks in my framework intersects with several strands of international research and policy on green jobs. It is useful to compare my task-based definitions with other prevalent approaches:

Output-based definitions. A common method, used by statistical agencies and early academic work, defines green jobs by the output of the firm or sector. The U.S. BLS, for example, in its Green Jobs Initiative (2010–2013) counted as green any job in an establishment that produces goods or services with an environmental benefit (renewable energy, pollution mitigation, recycling, etc.) (Elliott et al., 2024). However, this broad categorization also includes roles that may only indirectly contribute to sustainability, raising concerns about potential overestimation of green employment. Studies such as those by Vona et al. (2018) emphasize that many existing green job classifications rely too heavily on industry-level definitions, which fail to capture task-level environmental relevance. These limitations have prompted a growing body of work focusing on occupational and task-based metrics to understand the greenness of work more accurately. International Labour Organization (ILO) (2016) and OECD (2022) similarly argue for more granular approaches, emphasizing the need to measure not just whether a firm operates in a green industry, but whether individual tasks actually contribute to environmental objectives. In my study, I address this measurement gap by moving beyond output-based classifications to a task-level framework that better reflects the evolving content of work.. This “output approach” essentially designates entire industries or sub-industries as green and counts employment in them. Many national statistical offices in Europe followed a similar approach by measuring employment in the Environmental Goods and Services Sector (EGSS), which includes industries like renewable energy, waste management, and environmental consulting(Eurostat, 2016). While straightforward, the output-based approach has a binary character and can overlook green activities occurring in “non-green” industries. For example, a manufacturing company producing cars might implement cutting-edge energy-saving processes in its factories; output-based met-

rics would still count all its jobs as non-green because cars are not an environmental good per se.

Process-based and hybrid definitions. To capture environmentally beneficial activities outside the dedicated green sectors, some definitions include a *process perspective*. The BLS extended definition explicitly recognized this, adding a category for jobs involving environmentally friendly production processes and practices within any establishment (Vona et al., 2018) (Dierdorff et al., 2009). These could include roles like sustainability managers, energy auditors, or any workers tasked with improving a firm’s internal environmental performance (regardless of the firm’s main product). Sommers (2013) describes this as identifying establishments that use processes which “*conserve natural resources or reduce pollution.*” Internationally, the ILO also distinguished between employment in environmental *outputs* and employment in environmental *processes* when defining “employment in environmental activities” (Dierdorff et al., 2009). Moreover, the ILO introduced an additional criterion that a “green job” should be a decent job – i.e., providing adequate wages, rights, and social protection – arguing that sustainability has both environmental and social dimensions (Vona et al., 2018). Most statistical implementations (including ours) set aside the decent work criterion as it pertains more to job quality than environmental content.

My task-based approach can be seen as a refinement of the process-based perspective. Rather than classifying whole jobs as process-based green jobs (which could be ambiguous), I zoom in to identify which tasks within a job are related to environmental processes. In effect, I break each occupation into its constituent parts and evaluate those parts. This resonates with what some researchers call a “bottom-up” approach (Vona et al., 2018), identifying green activities at the micro level within firms or occupations, instead of labeling at the macro level of entire occupations or sectors.

Task-based approaches in research. A growing body of academic work has also employed task data to identify green jobs or the greenness of jobs. Early contributions in this vein include Dierdorff et al. (2009), who were among the first to systematically

incorporate the concept of green tasks. Using the O*NET database of occupational descriptors in the United States, they categorized occupations based on how the emerging green economy affects them. Specifically, they introduced categories of occupations such as *Green Increased Demand* (existing occupations that will see rising employment due to green trends), *Green Enhanced Skills* (existing occupations that will significantly change in task content and skill requirements due to greening, though the core work stays the same), and *Green New and Emerging* (entirely new occupations created by the green economy)(Vona et al., 2018). This taxonomy underscored that greening can reshape the task composition of jobs, even if the job itself does not vanish or newly appear.

Consoli et al. (2016) took a different angle by asking: in what ways do green jobs differ from others? Using U.S. O*NET data and employment records, they found that green jobs tend to require more formal education and high-level cognitive skills. Interestingly, they also found green jobs often demand greater *interpersonal* skills, hypothesizing that many green occupations (like environmental consulting, project management for sustainability initiatives) involve coordinating and communicating across disciplines, hence the need for teamwork and communication skills. Their results, published in *Research Policy*, suggested that green jobs are generally high-skilled and knowledge-intensive roles, dispelling a notion that green jobs might all be “blue-collar” installation work. At the same time, they did identify a spectrum: some green jobs are relatively low-skill (e.g., basic recycling sorters) but many are high-skill (engineers, managers). This interpretation is supported by Vona et al. (2018), who found only modest skill gaps between green and brown jobs, with green occupations requiring slightly more technical expertise. In addition, Bowen et al. (2018) emphasize that green jobs often exhibit greater task complexity, further reinforcing their generally knowledge-intensive nature.

Building on that, Vona et al. (2018) combined O*NET task measures with U.S. labor statistics to examine skill differences between green and brown occupations. They concluded that, contrary to some expectations, the overall skill gap between green and brown jobs is small(Consoli et al., 2016). Green jobs did not uniformly require dramatically different skills than brown jobs; rather, they mostly demanded some additional

technical competencies (for example, knowledge of environmental regulations or specific clean technologies) on top of a similar base of general skills(Vona et al., 2018). In other words, a worker in a brown occupation often has a skill set that is largely transferable to a green occupation, needing only targeted upskilling (Vona et al., 2018). This is an important insight for worker transition policies. Similarly, Vona et al. (2019) find that green employment in the U.S. is concentrated among higher-skilled, higher-wage workers and in certain regions, but again these green workers are not fundamentally a different breed—they are often drawn from adjacent occupations with overlapping skill requirements. Similarly, Vona et al. (2019) find that green employment in the U.S. is concentrated among higher-skilled, higher-wage workers and in certain regions, but again these green workers are not fundamentally a different breed—they are often drawn from adjacent occupations with overlapping skill requirements(Vona et al., 2019).

The GOJI approach inherits the spirit of these task-based analyses but with a dynamic twist: I apply it to a time series of occupational descriptions (in Germany) that are updated regularly. This allows me not just to classify which occupations are green or brown, but to observe how their task composition shifts year by year. Such within-occupation changes can be significant; indeed, Freeman et al. (2020) (analyzing U.S. data) found that changes in task content *within occupations* accounted for a major share of overall changes in job task requirements economy-wide over recent decades. In the context of environmental change, capturing within-occupation greening is crucial, since much of the transition involves existing jobs evolving rather than entirely new job titles appearing overnight.

Discussion of white (neutral) tasks. One might ask: if an occupation has mostly white tasks and a small fraction of green or brown tasks, should it be considered a green or brown job at all? This question highlights the nuance introduced by my continuous measure. Traditional approaches might force a cutoff (e.g., an occupation is labeled green if more than 50% of its tasks are green, or if it belongs to a green sector). In contrast, I embrace the continuum. Many occupations will indeed have GOJI values near zero—meaning they are largely neutral. That does not render the concept moot; rather, it

reflects reality. A large portion of the labor market (e.g., many service and administrative jobs) does not directly interact with environmental objectives. These jobs are “white” in my scheme. The GOJI allows me to identify which of those neutral occupations might be moving in a greener or browner direction. For example, consider a facilities manager: historically their tasks might all be neutral (managing building operations). If today that job starts to involve overseeing an energy efficiency retrofit (a green task), the GOJI will capture a shift from 0 to a small positive number. This granularity is valuable for pinpointing subtle diffusion of green practices across the economy.

Critique from a complementary perspective. A recent JRC working paper by Villani, GonzálezVázquez, and Fernández-Macías (2025) presents a compelling critique of occupational and task-based frameworks such as the GOJI. The authors argue that relying on occupational titles and predefined task lists can lead to both false positives (i.e., non-green jobs misclassified as green) and false negatives (i.e., environmentally relevant jobs not identified as green). These issues arise due to outdated classifications, inconsistent application across countries, and the erroneous assumption that task content remains stable over time (Villani et al., 2025). Based on empirical data from 24 European countries, they find that the share of green employment—when measured through task-based proxies such as O*NET—has not significantly increased over time and shows only a weak correlation with actual environmental outcomes. This suggests that task-based indicators may produce misleading conclusions if interpreted in isolation.

While the GOJI provides a continuous and occupation-level indicator of environmental task potential, it is not immune to the methodological challenges outlined by Villani et al. To mitigate these limitations, I employ several quality-assurance measures. First, the requirements catalog of the German BERUFENET system is continuously updated. New tasks are evaluated by a dedicated editorial team at the Federal Employment Agency, and environmentally relevant tasks are marked and visually highlighted using a green-leaf symbol. Second, all green and brown tasks undergo a keyword-based validation process at regular two-year intervals to ensure that classification remains accurate and internationally comparable. Third, where available, I triangulate GOJI scores with

firm- and sector-level environmental indicators—such as the presence of environmental management systems, environmental attitudes of firms, and the CO₂ intensity of sectors. Finally, I emphasize that GOJI values should be interpreted as potential indicators rather than definitive measures of ecological impact. They must be contextualized in relation to firm-level practices, sectoral dynamics, infrastructure, and the actions and attitudes of individual workers. These enhancements are intended to strengthen the interpretive validity of GOJI while addressing the concerns raised in the broader methodological debate.

In summary, the conceptual foundation of GOJI rests on viewing green labor through a task lens. This approach harmonizes with international calls to incorporate both output and process measures of green employment ([Apostel and Barslund, 2024](#), [Wang and Li, 2025](#)), and it operationalizes those ideas at the micro level. It moves beyond the dichotomy of “green job / not green job” and instead acknowledges shades of green (and brown) within each job. As the next section will show, this concept has been implemented in successive versions of my index, progressively refining how tasks are identified and quantified as green or brown.

3 Methodological Evolution of the GOJI Index

Developing a robust measure of the greenness of jobs has been an iterative process. In this section, I outline the history of the Greenness-of-Jobs Index, from its inception to the latest version (GOJI 2025). I highlight the rationale behind methodological changes and improvements at each stage. Three main stages can be distinguished: (1) the original GOJI (2018) which introduced the task-based greenness measure focusing solely on green tasks; (2) a beta version (circa 2021) that incorporated a notion of brown tasks with a cautious (initially rigid) definition; and (3) the current GOJI 2025, which features a refined and less restrictive classification of brown tasks along with other enhancements.

3.1 GOJI 2018: Pioneering a Task-Based Greenness Measure

The concept of GOJI was first operationalized by Janser (2018, 2019), who constructed an index to quantify the greenness of occupations in Germany. This inaugural version of GOJI was groundbreaking in that it applied text mining techniques to a comprehensive database of occupational task descriptions (the German *BERUFENET*, described in detail in Section 4) to identify environment-related content. The GOJI 2018 took a relatively straightforward form: for each occupation, it measured the share of that occupation’s tasks which were deemed “green,” i.e., related to environmental protection or resource conservation(OECD, 2023). In other words, if an occupation had T tasks in total and G of those were identified as green tasks, the index value was $GOJI = G/T$. This yielded a number between 0 and 1 (or 0 and 100 if expressed as a percentage) indicating the proportion of tasks that are green.

The creation of GOJI 2018 involved generating a dictionary of environmental keywords and phrases, and then scanning occupational descriptions for occurrences of those terms. These terms were drawn from domains such as renewable energy, energy efficiency, waste management, pollution control, and environmental regulation(Martinez-Fernandez et al., 2010)(International Labour Organization, 2018). For example, words corresponding to “solar,” “wind turbine,” “recycling,” “hazardous waste,” “emission control,” etc., were part of the search lexicon. If a task description contained any such keyword, that task was flagged as green.

One key feature of the original GOJI was that it implicitly treated all other tasks as non-green (effectively neutral). There was no separate accounting of negatively environmental (brown) tasks; occupations simply got credit for green content, and if they lacked any green keywords they would score 0. In practice, many occupations did register zero GOJI in early years, which is intuitive (plenty of jobs had no environmental tasks). Meanwhile, a few occupations scored relatively high. For instance, Janser (2019) reports that specialized environmental occupations such as “environmental protection officer” or “recycling engineer” had among the highest GOJI values, indicating that the majority of their tasks were related to environmental protection(Vona et al., 2019)(Janser, 2019).

In one example, the occupation “Specialist in environmental protection” had a GOJI of about 0.889, meaning roughly 89% of its listed tasks were green tasks(Janser, 2019). This makes sense for an occupation dedicated to environmental management.

At the other end, typical occupations in manufacturing or services that did not explicitly engage with environmental issues scored 0. The 2018 analysis revealed that there had been a modest upward trend in the average GOJI from 2006 to 2016 in Germany, suggesting an overall greening of jobs over that decade(Janser, 2019)(Janser, 2019). It also found variation across sectors and regions: certain industries like renewable energy and waste management unsurprisingly had high average GOJI, whereas others like traditional manufacturing or transportation had low values(Janser, 2019). Regional differences reflected the industrial composition and presence of green sectors in different areas.

While the original GOJI was a milestone, it had limitations that became apparent in subsequent analysis. One limitation was the exclusive focus on green tasks, treating all other tasks as equal in their non-greenness. This meant that an occupation engaged in environmentally harmful activities would only be reflected in the index indirectly (through an absence of green tasks, yielding a low score). For example, a coal mining technician job might score 0 because none of its tasks are environmentally friendly, but intuitively one might want to categorize that job as explicitly *brown* or polluting in nature. GOJI 2018 did not differentiate between a coal miner and, say, an office secretary—both could have a zero score if they have no green tasks, despite the former having strongly negative environmental implications and the latter being neutral.

Another limitation was methodological: the identification of tasks relied on keyword matching and weighting through a content analysis approach described as lexicometric. There was an implicit assumption that the presence of environmental keywords adequately captures task greenness, and that each task is either green or not. Context and nuance were not deeply addressed; for instance, a task described as “manage emissions from industrial process” might have been counted as green (because it involves emissions management, presumably to reduce pollution), whereas a task “operate coal furnace” would simply not trigger any green keyword and thus be neutral by omission.

Despite these issues, GOJI 2018 provided proof of concept that task-based greenness can be measured and can yield meaningful aggregate insights. It showed positive correlations between an occupation’s greenness and its employment growth in Germany(Janser, 2019), hinting that greener jobs were on the rise and expanding faster than less green ones. These findings motivated further refinement of the index.

3.2 Incorporating Brown Tasks: GOJI Beta Version

After the initial publication, efforts were made to improve GOJI by explicitly accounting for *brown tasks*—those tasks that contribute to environmental harm. The next iteration, which I term a beta version (developed in the early 2020s), introduced a negative component to the index. The idea was to construct not one but two measures: a **GOJI⁺** (positive greenness index) akin to the original GOJI measuring green task share, and a **GOJI⁻** (negative greenness index) measuring the share of brown tasks in an occupation. This development was influenced by a recognition that policy and research are interested not only in how green jobs are, but also how “brown” certain jobs are, especially in the context of transitioning away from polluting industries.

In implementing the brown task measure, the initial approach was deliberately cautious or “rigid.” A rigorous definition of brown tasks was used to avoid false positives (i.e., incorrectly labeling tasks as brown when they might not truly be environmentally harmful). The team compiled a list of keywords that signify environmentally detrimental processes and technologies, focusing on the obvious culprits: words related to fossil fuels (e.g. “coal,” “oil,” “diesel engine,” “gasoline,” “natural gas”), high-emission industrial processes (“blast furnace,” “boiler combustion,” “incinerator”), heavy use of energy or materials (“smelting,” “petrochemical,” “plastic production”), and known pollutants (“pesticides,” “solvents,” etc.). Only tasks containing these kinds of terms would be flagged as brown.

This initial brown dictionary was purposefully narrower than it could have been, to ensure that tasks were not mislabeled. For example, consider a task description like “operate diesel generator for backup power.” That clearly involves fossil fuel use and

would be counted as a brown task. But a task like “drive a delivery truck” might not explicitly mention diesel or fuel, even though it likely involves emissions. The rigid definition in the beta phase might miss such cases because it relied on explicit textual cues. In the beta version, if the task description didn’t include a clear brown keyword, it remained unflagged (neutral) even if common sense tells me the task has environmental negatives.

At the same time, the beta version maintained essentially the same methodology for green tasks as GOJI 2018, possibly with minor updates to the green keyword list (some new terms for emerging technologies, etc., might have been added as the landscape evolved — for instance, more mentions of “electric vehicle” or “LED lighting” tasks by the early 2020s).

With both GOJI^+ and GOJI^- in hand, a combined index was defined. The **net GOJI** for an occupation could be calculated as GOJI^+ minus GOJI^- . This net index gives a single number that increases with more green task intensity and decreases with more brown task intensity. An occupation performing both green and brown tasks could have those effects offset in the net value. (I formally define these measures in Section 4.2.) The introduction of a net measure allowed occupations to be ranked on a spectrum from very brown (negative net values) through neutral (around zero) to very green (positive net values).

Researchers used this enhanced index in exploratory analyses. An example outcome from that period is the classification of occupations into categories like “dark green,” “light green,” “neutral,” “light brown,” and “dark brown” based on threshold ranges of the net GOJI. One scheme (later adopted in the GOJI 2025 analysis) used cutoffs around ± 0.1 of the net index (Bachmann et al., 2024). Occupations with net GOJI above $+0.1$ were labeled *dark green* (meaning they have substantially more green task content than brown, relative to all tasks), those between 0 and $+0.1$ were *light green*, those exactly 0 or near 0 were neutral, and similarly on the negative side for brown. Using data for year 2012 as an example, it was found that only a small fraction of occupations fell into the extreme categories: about 7% were dark green and 6% dark brown, whereas roughly half

(51%) were neutral and the remainder were light green or light brown (Bachmann et al., 2024). This intuitive result indicated that while truly green-intensive or brown-intensive occupations exist, most jobs have relatively low intensity of either kind of environmental tasks.

During the validation of the beta version, it became clear that some task descriptions were too ambiguous for a consistently narrow definition of brown tasks. Additionally, a few false negatives were identified—tasks clearly linked to fossil technologies that had been missed. These were added to GOJI v2 to improve coverage.

This insight led the team to adjust the strategy for GOJI 2025, making the brown task identification less rigid (more inclusive) while still aiming for accuracy.

3.3 GOJI 2025: Refinement and Finalization

The current version, GOJI 2025, builds upon the beta experience to provide a more comprehensive and balanced measure of occupational greenness. Several enhancements characterize GOJI 2025:

- **Expanded and refined keyword dictionaries:** The list of green and brown keywords was revisited and expanded. On the green side, new terms reflecting recent technological and policy developments were added (for example, terms related to electric vehicles, battery technology, energy auditing, circular economy practices, etc., which have grown in importance since 2018). On the brown side, the dictionary was broadened beyond the narrow initial set. During this refinement, the team analyzed task descriptions of known brown-intensive occupations to identify commonly used phrases that should count as brown. This led to including more general terms like “internal combustion engine,” “fuel consumption,” “emissions test” (if in context of producing emissions rather than reducing them), and so forth. Essentially, the brown classification criteria became less rigid by capturing a wider array of language that implies environmental harm. The total number of distinct keyword patterns in GOJI 2025 is sizable: as documented, around 399 keywords/phrases are used for green tasks and 191 for brown tasks. These figures underscore the

thoroughness of the classification scheme.

- **Improved text-processing (NLP) techniques:** While still fundamentally based on pattern matching, GOJI 2025 employs more systematic use of regular expressions and text normalization. Regular expressions allow for matching variations of words (e.g., singular/plural, different verb tenses) and accounting for word boundaries (to avoid false matches where a substring appears coincidentally in an unrelated word). For example, the pattern ‘ ‘| ’ ’ would catch “Solartechnologie” or “Solaranlage” in German text, capturing references to solar technology; patterns can also allow prefixes and suffixes (e.g., “**klima**” to catch “Klimaschutz” (climate protection) or “klimafreundlich” (climate-friendly)). GOJI 2025’s methodology, described in Appendix A.2, involves a two-step matching: first identify all task statements containing any of the green or brown keywords, then verify and categorize them accordingly. The result is a set of flagged green tasks $G_{o,t}$ and brown tasks $B_{o,t}$ for each occupation o in year t .
- **Explicit neutral baseline:** Since all tasks not flagged as green or brown are considered neutral, GOJI 2025 simply computes the total tasks $T_{o,t}$ for each occupation from the data (combining core and additional tasks). By using the complete set of tasks as a denominator, the index inherently accounts for how “important” the environmental tasks are relative to the occupation’s total job duties. If an occupation expands its list of tasks over time (say new tasks are added in descriptions), the denominator grows, which can dilute the shares of green or brown if those don’t grow proportionally. This ensures that an occupation doesn’t appear greener just because it added more detailed descriptions of its tasks; it only appears greener if it added more *green-related* content.
- **Net index and category thresholds:** GOJI 2025 retains the net index concept (I often call it simply “the GOJI” as a singular measure now, with the understanding that it embodies both positive and negative components). After computing

$GOJI_{o,t}^+ = G_{o,t}/T_{o,t}$ and $GOJI_{o,t}^- = B_{o,t}/T_{o,t}$, I define:

$$GOJI_{o,t} = GOJI_{o,t}^+ - GOJI_{o,t}^-, \quad (1)$$

which can range theoretically from -1 (if all tasks are brown) to $+1$ (if all tasks are green). In practice, most values are in a narrower range. Empirically, I observe the lowest occupational values around -0.4 and highest around $+0.8$ in the data (for instance, one of the greenest occupations in recent data is environmental engineers, and one of the brownest might be coal mining machine operators). GOJI 2025 also formally adopts the threshold scheme introduced in the beta: I label occupations as “dark brown” if $GOJI < -0.10$, “light brown” if $-0.10 \leq GOJI < 0$, “neutral” if $GOJI = 0$ (or very close to 0 in practice, up to ± 0.005 say), “light green” if $0 < GOJI \leq 0.10$, and “dark green” if $GOJI > 0.10$. These thresholds correspond roughly to the average shares of brown and green tasks across occupations (about 10% each). This categorization is useful for communication and analysis, though researchers can always use the continuous index for more precision.

- **Annual updates and historical coverage:** GOJI 2025 is computed on occupational data from 2011 through 2022 (and will be extendable beyond as data become available). The choice of starting at 2011 was due to a change in the occupational classification in Germany (the KldB 2010 system was implemented around that time). The original GOJI had also used data from 2006 and then 2011 onward; GOJI 2025 ensures consistency by focusing on a continuous period under one classification scheme, avoiding a structural break in the data. Having a decade-plus of yearly observations allows me to observe trends. Importantly, occupational descriptions in BERUFENET are updated asynchronously – not all occupations get new descriptions every year, but enough do that changes can be tracked. GOJI 2025 takes each year’s snapshot of descriptions and computes the index anew. As a result, it can capture both the emergence of new green tasks in occupations and the phasing out of brown tasks over time. For example, between 2012 and 2022, many

technical occupations began to mention working with renewable energy or energy efficiency, and some heavy industrial jobs reduced emphasis on certain polluting tasks (or those tasks were de-emphasized in descriptions as those aspects decline).

- **Validation and examples:** In developing GOJI 2025, more effort was put into validation. This involved manually examining a sample of occupations across the green-brown spectrum to see if their index values “made sense” given external knowledge. Encouragingly, occupations like solar installers, recycling specialists, and environmental auditors came out with strongly positive GOJI, whereas occupations like oil refinery operators, coal miners, and diesel mechanics had negative GOJI values, as expected. Occupations that one would intuitively consider neutral (like primary school teachers, accountants, retail cashiers) mostly registered GOJI = 0, having no task hits in either category. There were a few borderline cases that prompted adjustments to the dictionaries. For instance, early runs showed a certain technical occupation had a moderate negative GOJI; on inspection, it turned out the occupation’s description included a term like “Abgas” (exhaust gas) in the context of *reducing* emissions. The algorithm had wrongly flagged it as brown due to the word “exhaust gas.” To fix such cases, context-specific rules or additional green phrases (like “emission reduction”) were added, so that tasks explicitly about reducing or managing emissions are counted as green, not brown.
- **Rigid, but balanced:** I describe the brown task definition in GOJI 2025 as more “rigid” than before, meaning it casts a narrower net. But it is still carefully curated to avoid undercounting. The aim was balance: both $\text{GOJI}^{\text{green}}$ and $\text{GOJI}^{\text{brown}}$ should meaningfully reflect the presence of environmentally relevant tasks. If the criteria are too loose, I might erroneously flag tasks as green/brown; if too strict, I miss legitimate cases. In GOJI 2025’s classification very few occupations have both a high green and a high brown share simultaneously. Almost all occupations which are high on one tend to be low on the other. This suggests that “cancelling out” in the net index is minimal; occupations generally skew one way or are neutral. Intuitively, a job that heavily involves fossil-fuel tasks usually doesn’t also involve

a lot of eco-friendly tasks, and vice versa, so treating the net as a summary is reasonable.

Overall, GOJI 2025 represents a mature version of the index, one that has incorporated lessons from earlier versions and feedback from its application in research (including an analysis of labor market flows using the index, as referenced later). It provides a rich portrayal of how green or brown each occupation is and how that changes over time.

To illustrate the evolution with a concrete example, consider the occupation of *Automotive Mechanic*. In the 2018 GOJI, this occupation likely had a GOJI of 0 (no green task detected, since tasks revolve around repairing engines, brakes, etc., none explicitly green). The beta version introduced brown tasks: an automotive mechanic’s description might mention working with combustion engines and exhaust systems, so perhaps one or two brown keywords would trigger. If, say, 1 out of 10 tasks was flagged brown, $GOJI^- = 0.1$ and $GOJI^+ = 0$, net $GOJI = -0.1$, classifying it as light brown. However, if the description never explicitly said “fuel” or “emissions,” maybe even that was missed in the beta. By GOJI 2025, the brown lexicon is broader, possibly catching terms like “internal combustion engine maintenance” or “exhaust testing” that are common in such job descriptions. So now the mechanic might register a brown task or two. Meanwhile, newer automotive technology has led some mechanics to have tasks like “service electric vehicle systems” in their profile – a new, greenish task. If such a task is present, it would count as green. The net outcome for *Automotive Mechanic* in 2022 might be a GOJI very close to zero: a bit of brown content balanced by a bit of green content. This nuanced result aligns with reality: auto mechanics are in a greening field (with EVs) but still dealing mostly with gasoline engines.

In summary, the path from GOJI 2018 to GOJI 2025 has moved from a simple green-task share to a more sophisticated index capturing both sides of the coin. I now have a tool that not only measures how green occupations are, but also how brown they remain, providing a fuller picture of the challenge in greening the labor market. The next section will delve into the data sources and the detailed methodology used in GOJI 2025, demonstrating exactly how tasks are coded and the index is calculated.

4 Data and Coding Methodology for GOJI 2025

In this section, I describe the data source and the technical steps involved in constructing the Greenness-of-Jobs Index (GOJI) 2025. I start with an overview of the BERUFENET occupational database and how it provides task information. I then detail the text-mining procedure used to identify green and brown tasks, including illustrative examples of task coding. Finally, I present the formulae for computing the index and discuss some examples of occupations to demonstrate the calculation.

4.1 Data Source: BERUFENET Occupational Descriptions

My analysis relies on textual descriptions of occupations from **BERUFENET**, which is an online database of occupations maintained by the German Federal Employment Agency. BERUFENET is essentially a comprehensive repository of job profiles, containing information such as job titles, brief descriptions, typical tasks and duties, required qualifications, knowledge and skills, etc. For the purposes of GOJI, the key component of BERUFENET is the *requirements catalog* (“Anforderungskatalog”) of each occupation, which lists the tasks and activities associated with that occupation.

Note on the interpretation of task classification. It is important to emphasize that the GOJI classification reflects the *potential* environmental impact of occupational tasks. The actual ecological effect of a job depends on multiple contextual factors, including the environmental footprint of the firm or sector, the nature of the goods and services produced, the energy and resource efficiency of production processes, the means of transport and logistics used, the sustainability of regional infrastructure, and finally the personal attitudes and actions of individual workers. A task may be coded as ‘green’ because it involves environmentally relevant objectives (e.g., energy management), but its impact will vary significantly depending on how it is implemented in practice.

Some important characteristics of the BERUFENET data:

- **Occupational classification:** Occupations in BERUFENET are categorized according to the German Classification of Occupations (Klassifikation der Berufe,

KldB). I focus on the 2010 KldB system, which organizes jobs in a hierarchical 5-digit code structure. A 5-digit KldB code (e.g., “26402” for *Solar energy technician*, as a hypothetical example) represents a specific occupational grouping often referred to as an “Einzelberuf” (single occupation). There are over 4,000 such 5-digit occupations in the database. These are fairly granular (more detailed than the 3-digit or 2-digit broad groups).

- **Task descriptions:** Each occupation entry typically has a set of bullet points or short paragraphs describing core tasks (main duties essential to the job) and additional tasks (supplementary duties that some job-holders might perform). For example, a mechanical engineer occupation might have core tasks like “Design mechanical components and assemblies” and additional tasks like “Conduct energy-efficiency assessments of mechanical systems” (if relevant). BERUFENET distinguishes core vs. additional tasks in structure, but for my analysis I combine them because any task can potentially be green or brown. I do, however, keep track of the total number of tasks by summing core and additional.
- **Updates over time:** BERUFENET is a living database; the Federal Employment Agency updates occupational descriptions to reflect new technologies, changing work practices, and emerging job roles. Not all occupations are updated every year, but many will be revised every few years or when significant changes occur. New occupations can be added and obsolete ones may be removed or merged. In my dataset, I extract snapshots of the full set of occupational descriptions for each year from 2011 through 2022. By doing so, I capture how descriptions (and thus task content) evolve. For consistency, I track occupation codes that exist in a given year and measure their greenness that year. If an occupation is discontinued or merged, I handle that by its absence or mapping to a new code.
- **Language:** The descriptions are in German, as they are intended for career counselors, job seekers, and employers in Germany. My text-mining is performed on the German text. The examples I provide in this paper may be translated to English

for clarity, but it should be understood that the pattern matching was applied to German keywords. This is an important point: the specificity and nuance of the keyword lists are tailored to German terminology (including technical jargon and compound words often used in German occupational language).

- **Descriptive statistics:** Each annual snapshot contains roughly 3,000 to 4,500 occupations (the number grew over time as new occupations emerged, like those related to IT and green tech). Each occupation listing contains anywhere from a couple to a few dozen task items. In total, the 2020 snapshot had about 7,671 distinct task statements across 4,422 occupations (Kuai et al., 2025), meaning an average of 1.7 unique tasks per occupation. However, because tasks are often reused (the same task description may appear under multiple related occupations), the median number of tasks per single occupation entry might be higher when counting duplicates. In practice, an occupation entry might list 5-10 tasks, but many of those tasks are generic and appear in other entries too.

The richness of this data allows a fine-grained measure like GOJI to be feasible. Without such a database, one would have to rely on occupational titles or sector classifications as proxies for greenness, which are far less precise. For example, the BERUFENET description for *Gärtner/in - Garten- und Landschaftsbau* (Gardener - Landscaping) includes tasks such as “Plan and construct sustainable green spaces” (which might be flagged as green if it includes words like “sustainable”) and “Operate construction machinery for earthworks” (which might be neutral or brown depending on wording). This level of detail is what I exploit.

A note on scope: GOJI is computed for occupations, not individual jobs or workers. It does not tell me which specific worker is in a green job, but rather how green the typical job in a given occupation is. To relate GOJI to actual employment, one can combine it with employment data by occupation (which I do in some analyses, for instance computing the share of total employment that is in green tasks, etc.). But the construction of GOJI itself is purely on the occupation description side.

4.2 Identifying Green and Brown Tasks: Text-Mining Procedure

The crux of the GOJI methodology is identifying which tasks in the BERUFENET descriptions count as green or brown. I achieved this through a text-mining algorithm using a predefined set of keywords and phrases (regular expressions) for each category.

The procedure can be summarized in steps:

1. **Preprocessing text:** All task descriptions are cleaned and standardized. This involves converting text to lowercase (since the keyword matching is case-insensitive), removing any irrelevant characters or markup, and perhaps replacing synonyms or abbreviations with a standard form (for example, if “CO₂” appears, I ensure the keywords can catch it as “CO2” or “Kohlendioxid” or similar; if “etc.” appears in text, it’s ignored, etc.). German umlauts and special characters are also handled to ensure matching (e.g., “Ö” becomes “oe” for matching if necessary, or the patterns account for them).

2. **Green keyword matching (first pass):** I apply the list of green regex patterns to each task description. Each pattern is designed to detect a particular concept.

For example:

- Phrases relating to renewable energy: `\Solar"`, `\Windkraft"`, `\Photovoltaik"`, `\Biomasse"`, etc.
- Energy efficiency terms: `\Energieeinspar"` (to catch words like `Energieeinsparung`, energy saving), `\Wärmedämm"` (insulation), `\energetische Sanierung"` (energy retrofit).
- Environmental protection processes: `\Umweltschutz"` (environmental protection), `\Emissionsminderung"` (emission reduction), `\Abfallvermeidung"` (waste avoidance).
- Environmental technology: `\Kläranlage"` (wastewater treatment plant), `\Filteranlage"` (filtration system).

- Green certification and compliance: `\Umweltaudit`", `\ISO 14001`", `\Energiemanagement`".
- Miscellaneous sustainable practices: `\nachhaltig`" (sustainable) if tied to a task like procurement, `\umweltfreundlich`" (environmentally friendly), `\recyceln`"/`\Recyc`
`\kompostieren`" (composting), etc.

If a task description matches any green pattern, it is marked as a *green task*. I store which occupation it belongs to and increment that occupation's green task count $G_{o,t}$.

3. **Brown keyword matching (first pass):** Similarly, I apply the brown list:

- Fossil fuel related: `\Kohle`" (coal), `\Erdöl`"/`\Öl`" (oil), `\Erdgas`" (natural gas), `\Benzin`" (gasoline), `\Diesel`", `\Kraftstoff`" (fuel).
- Emission processes: `\Abgas`" (exhaust gas), `\Rauchgas`" (flue gas), `\Emission(?!sarm)`" (the regex might exclude “emission-free” or “emissionsarm” which means low-emission, to not misclassify a green context as brown). This keyword-based task classification approach is informed by international standards. The ILO and Cedefop have previously developed a structured taxonomy for identifying green skills and occupations based on detailed task descriptors ([International Labour Organization and European Centre for the Development of Vocational Training \(Cedefop\), 2012](#)). Such taxonomies provide a foundation for mapping job content to environmental relevance and are widely used in policy research and labor statistics.
- Energy-intensive equipment: `\Ofen`"/`\Hochofen`" (furnace/blast furnace), `\Verbrennung`" (combustion), `\Kessel`" (boiler), `\Turbine`" (if specifically gas/steam turbine context, though “Turbine” could be wind too; context matters).
- Polluting materials and chemicals: `\Dieselmotor`" (diesel engine), `\Otto-Motor`" (gasoline engine), `\Asbest`" (asbestos), `\Schwermetall`" (heavy metal), `\Pestizid`", `\Lösemittel`" (solvents).
- Carbon-intensive production: `\Stahlherstellung`" (steel production), `\Zement`"

(cement, since cement production is very CO₂-intensive), \Kraftwerk" (power plant, contextually fossil).

- Environmental harm keywords: \Umweltbelastung" (environmental burden), \entforsten" (deforest, not sure if present in tasks, but possibly in forestry context as negative).

If a task matches a brown pattern, it's marked as a *brown task* and I increment $B_{o,t}$ for that occupation.

4. **Avoid double counting conflicts:** It is possible (though uncommon) that a task could contain both green and brown keywords. For example: "Überwachen von Emissionen und Energieverbrauch, um Einsparungen zu erzielen" ("Monitor emissions and energy consumption to achieve savings"). Here the word "Emissionen" might trigger a brown flag, but the context is actually about reduction (savings). I have designed the patterns to mitigate this—for instance, by requiring certain context or excluding terms like "reduzieren" (reduce) from brown triggers. In cases where a task nonetheless matches both sets (due to overlapping patterns), I decide a priority. Typically, I would classify it as a green task if it contains any explicit indication of mitigation or improvement (the rationale being that the task's intent is environmental improvement, even if it deals with a pollutant). In code, I might implement that by running green matches first, then not counting those tasks as brown even if a brown keyword appears, or by more refined pattern definitions. Ultimately, every task should be counted at most once: either green, or brown, or neutral.

5. **Calculate shares:** For each occupation o in year t , after processing all its tasks, I have:

$G_{o,t}$ = number of green tasks identified in occupation o ,

$B_{o,t}$ = number of brown tasks identified in occupation o ,

$T_{o,t}$ = total number of tasks listed for occupation o (core + additional).

I then compute the positive and negative greenness-of-jobs indices:

$$GOJI_{o,t}^+ = \frac{G_{o,t}}{T_{o,t}}, \quad (2)$$

$$GOJI_{o,t}^- = \frac{B_{o,t}}{T_{o,t}}. \quad (3)$$

The net greenness index is:

$$GOJI_{o,t} = GOJI_{o,t}^+ - GOJI_{o,t}^-.$$

This net index can also be expressed as $(G_{o,t} - B_{o,t})/T_{o,t}$, which is effectively the difference in the proportion of tasks that are green versus brown. An equivalent interpretation: $GOJI_{o,t}$ is the share of tasks that are green minus the share that are brown.

To ensure clarity, let’s work through an **illustrative example**. Consider the occupation *Heating, Ventilation, and Air Conditioning (HVAC) Installer* (in German, roughly *Anlagenmechaniker für Sanitär-, Heizungs- und Klimatechnik*). In the year 2022, suppose BERUFENET lists 8 tasks for this occupation:

[i)] Install and maintain heating systems (gas and oil boilers). Install and maintain solar thermal panels for water heating. Adjust and repair ventilation and air conditioning systems. Optimize energy efficiency of heating systems through control settings. Dispose of old heating units and refrigerants according to regulations. Advise customers on energy-saving options and renewable alternatives. Assemble pipe systems and connect radiators. Ensure compliance with environmental standards in HVAC installations.

Now, let’s apply the coding: - Task (i) mentions *gas and oil boilers*. Keywords: “Gas” and “Öl” in context of boilers. Likely the brown list catches “Ölkessel” or “Heizkessel” (boiler) if present. This task is clearly about fossil fuel heating – I would count it as a brown task (due to oil/gas boiler). - Task (ii) mentions *solar thermal panels*. The word “Solar” should be in the green list. This is a classic green task (renewable energy

installation). - Task (iii) is about ventilation/AC maintenance – probably neutral, as it’s neither inherently green nor explicitly brown (though AC refrigerants can be an environmental issue, the text doesn’t say anything like “CFC” or so; if it did say “handling of refrigerants” maybe brown due to chemicals, but let’s assume neutral phrasing). - Task (iv) has *Optimize energy efficiency*. “Energieeffizienz” or similar would be a green trigger. So that’s a green task. - Task (v) *Dispose... according to regulations* – this is borderline: it’s dealing with hazardous materials (old units, refrigerants). Could be partly environmental in that it’s proper disposal (which is environmental protection), or one could see handling refrigerants as a brown element if releasing them is harmful. However, “according to regulations” implies an environmentally responsible practice. I likely have “Entsorgung” (disposal) in green list especially if tied to regulations or recycling. I’ll assume it’s considered a green task (because it’s managing waste properly). - Task (vi) *Advise on energy-saving and renewables* – definitely green (energy-saving, renewables mentioned). - Task (vii) *Assemble pipe systems...* – neutral (standard skill, no env context). - Task (viii) *Ensure compliance with environmental standards* – green (explicit environmental aspect).

Counting: Out of 8 tasks, green tasks: (ii), (iv), (v), (vi), (viii) → that’s 5 green tasks. Brown tasks: (i) likely one due to fossil boilers. Perhaps also (iii) if AC refrigerant was considered, but I said neutral. So $B = 1$. Thus $G_{o,2022} = 5$, $B_{o,2022} = 1$, $T_{o,2022} = 8$. Then $GOJI^+ = 5/8 = 0.625$, $GOJI^- = 1/8 = 0.125$, net $GOJI = 0.500$. This hypothetical HVAC installer ends up quite green-leaning, which might be plausible by 2022 if the job indeed heavily involves renewables and efficiency now. If I categorize: net 0.500 $\notin$ 0.1 so “dark green” occupation. (In earlier years, the same occupation might have been more brown: in 2010 maybe very little solar, mostly oil/gas work, so it could have been net brown then. GOJI can capture that shift.)

For a contrasting example, consider *Coal Mining Machine Operator*. Typical tasks: operate cutting machines in coal seams, transport coal, maintain machinery, ensure safety, etc. Green tasks: likely none (unless something about land reclamation which is unlikely in the job description). Brown tasks: many (cutting coal, handling coal, etc. would have

triggers like “Kohle”). So maybe $G = 0$, $B = 4$ out of 7 tasks, giving GOJI net around -0.57 . Very brown (dark brown category).

One more: *Office Clerk*. Tasks: file documents, use office software, schedule meetings, etc. None green or brown. So $G = 0$, $B = 0$, $T = say5$. $GOJI^+ = 0$, $GOJI^- = 0$, net 0. Neutral.

These examples highlight how GOJI works at the occupation level. Of course, not every occupation’s tasks are as clearly worded. Some require interpretation, which is why the careful crafting of regex patterns and iterative refinement (with manual checks) is vital.

4.3 Calculation of the Index and Aggregation

Once $GOJI_{o,t}$ is computed for all occupations o , I can aggregate or analyze the data in various ways for research or policy insights:

- I can compute the distribution of GOJI across occupations in a given year. As mentioned, it tends to cluster around zero with long tails. For instance, Figure 11 in Appendix A.2 (described in the source) shows a histogram of occupational GOJI in 2012(Consoli et al., 2016)(Vona et al., 2019), highlighting the percentage in each category (dark brown, light brown, neutral, etc.).
- I can also weight the occupational index by employment to get an economy-wide “greenness of employment” measure. Janser (2018, 2019) did this, finding an increase in the employment-weighted greenness index over time. Essentially, that approach multiplies each occupation’s GOJI by its share of employment and sums, or similar. This tells me what fraction of tasks being done *in the economy* are green (versus brown).
- The index can be merged with other data at occupational level, e.g. administrative registers at individual level, such as average wages, education levels, or demographic composition of occupations, to study correlations. For example, do greener occupations tend to pay higher wages? Several studies suggest so. Bowen et al. (2018)

find a modest green wage premium, particularly in high-skill roles. This is further supported by Kuai et al. (2025), who estimate a significant green wage differential using microdata from multiple countries. Regarding education, Consoli et al. (2016) report that green jobs often require more formal education and cognitive skills than non-green jobs. In terms of gender representation, Vona et al. (2019) highlight that green jobs are disproportionately held by men, raising concerns about gender equity in the green transition.

- I can identify which occupations are greening or browning over time by looking at $\Delta GOJI_o = GOJI_{o,2022} - GOJI_{o,2012}$, for example. A positive change means the occupation added green content or dropped brown content (or both). Preliminary analysis suggests that many technical and engineering occupations saw an uptick in GOJI, whereas some extraction and traditional manufacturing jobs remained flat or even increased in brown share (though many of those are declining in employment).

It is important to stress that GOJI is a measure of *potential environmental task content* as described in occupational profiles. It does not directly measure environmental performance or outcomes. For instance, if an occupation has a green task "promote recycling in the company," GOJI counts that, but whether every worker in that occupation actually does it effectively is another matter. The assumption is that the inclusion of the task in the profile indicates that this is part of the expected role, and presumably many job-holders will perform it. Similarly, if a brown task is listed, it indicates that job involves that harmful activity typically.

Finally, one might ask how GOJI deals with *new occupations*. For example, with the rise of the renewable energy sector, entirely new jobs like "wind turbine service technician" emerged in the 2010s. These occupations might not exist in 2012 data but appear by 2022. GOJI simply scores them when they appear. Often, they will have high GOJI (wind technician would have many green tasks, few or no brown). So the overall greenness of the labor market can increase both because existing jobs get greener and because new green jobs are created (and likewise, greenness can decrease if new brown jobs appear or if certain jobs get browner, though the latter is less common in recent

times). GOJI can capture both dynamics.

In conclusion, the data and methodology underpinning GOJI 2025 allow for a detailed measurement of the greenness and browning of occupations. With the occupation-level GOJI data computed, I now have a foundation to explore trends and patterns. Before turning to that analysis, I will review how my work connects with the broader literature.

5 Previous Research and Literature Review

Interest in “green jobs” and the labor market implications of environmental sustainability has grown substantially over the past two decades. In this section, I situate the Greenness-of-Jobs Index within the context of existing research. I review the literature along several dimensions: (1) efforts to define and measure green jobs, (2) comparisons of green vs. non-green jobs in terms of skills and wages, (3) empirical findings on employment impacts of the green transition at both micro and macro levels, and (4) studies of worker reallocation and transition from brown to green jobs. This review draws from a wide range of sources, including economics, environmental studies, and policy analysis, reflecting publications in journals such as *Technological Forecasting and Social Change*, *Ecological Economics*, *Journal of Labor Economics*, *Labour Economics*, *Research Policy*, *Journal of the Association of Environmental and Resource Economists (JAERE)*, *Journal of Environmental Economics and Management (JEEM)*, and others, as well as influential working papers (IZA, NBER, OECD, etc.).

5.1 Defining and Measuring Green Jobs

One of the earliest challenges was simply agreeing on what constitutes a “green job.” The debate has largely revolved around whether to use an output-based or process-based definition, as discussed in Section 2. Early 2000s work like [Rennings and Zwick \(2002\)](#) and policy initiatives by the ILO and UNEP laid conceptual groundwork, often focusing on employment in environmental sectors (renewable energy, energy efficiency industries, etc.). However, as [Deschenes \(2013\)](#) notes in his overview, estimates of green

employment can vary widely depending on definitions and measurement approaches. Deschenes highlighted a mixed approach: combining sector definitions (like the System of Environmental-Economic Accounting, SEEA, classification of environmental goods and services) with occupational information(Janser, 2019).

The U.S. BLS’s foray into measuring green jobs around 2010-2012 was a pivotal effort. BLS carried out surveys to count green jobs by two definitions: the output approach (the Green Goods and Services survey) and the process approach (a special employer survey on environmental practices). The results, summarized by Sommers (2013) (a BLS report), indicated that the green goods and services sector accounted for about 2-3

In Europe, statistical agencies took similar approaches to estimate the size of the Environmental Goods and Services Sector (EGSS) and the number of jobs therein. Horbach and Janser (2016) exemplify a research study using establishment survey data to identify firms offering green products or processes and comparing their employment growth. They found that German establishments active in green markets had higher employment growth during certain periods (2006-2012) than other firms, suggesting a growth potential in green activities(Janser, 2019)(Peters, 2014). However, that study also pointed out concentration of such jobs in specific regions and skill segments.

The limitations of broad definitions led to increased interest in task-based measurement. I have already covered how Dierdorff et al. (2009) introduced occupational categories impacted by the green economy. Their findings, while qualitative, indicated that few truly new occupations were created initially, but many existing ones would require new knowledge or skills (their “enhanced skills” category). This was echoed by Peters (2014), who quantitatively identified 176 occupations (out of 1,000 in the U.S.) that utilized green tasks to some degree. Peters’ analysis, published in an applied economics journal, was among the first to give a percentage of occupations engaged in the green economy (roughly 20%). He also noted that these occupations were spread across diverse industries, not just in traditional environmental sectors(Bluedorn et al., 2023). This reinforced the idea that the green transition permeates many corners of the labor market.

Consoli et al. (2016) took a different angle by asking: in what ways do green jobs

differ from others? Using U.S. O*NET data and employment records, they found that green jobs tend to require more formal education and high-level cognitive skills(Blueborn et al., 2023). Interestingly, they also found green jobs often demand greater *interpersonal* skills, hypothesizing that many green occupations (like environmental consulting, project management for sustainability initiatives) involve coordinating and communicating across disciplines, hence the need for teamwork and communication skills. Their results, published in *Research Policy*, suggested that green jobs are generally high-skilled and knowledge-intensive roles, dispelling a notion that green jobs might all be “blue-collar” installation work. At the same time, they did identify a spectrum: some green jobs are relatively low-skill (e.g., basic recycling sorters) but many are high-skill (engineers, managers).

Following on that, Vona et al. (2018) examined whether the skill gap between green and brown jobs might hinder workers from transitioning between them. Using the O*NET classification of green tasks and merging with wage and employment data, their JAERE paper found that on average, a brown occupation and a green occupation have very similar skill profiles except that the green one requires an extra dose of specific technical knowledge. For example, an electrician (brown if working on coal power plant wiring) versus a solar PV installer (green): both need electrical skills, but the latter needs some knowledge of solar technology. Thus, re-skilling might be easier than presumed, an encouraging conclusion for training policy. They did caution, however, that certain technical skills (like knowledge of environmental regulations or specific renewable tech) are not yet widespread, which could create short-term bottlenecks.

Vona et al. (2019) extended this to look at spatial and distributional aspects in the U.S. They confirmed that green jobs were concentrated in particular regions (often aligning with places that had strong environmental policies or tech sectors) and that individuals in green jobs on average earned more, were more educated, and were more concentrated among males (pointing to a possible gender gap in green employment). This work, appearing in *Journal of Economic Geography*, stresses that the rise of green jobs might contribute to new regional inequalities if not balanced (since some areas might lose brown

jobs while others gain green jobs).

From a measurement standpoint, the GOJI approach aligns strongly with these task-based efforts. The innovation is applying it to a new country (Germany) with rich data and doing it dynamically over time. In a sense, GOJI is a measure of the “greening of occupations” akin to what others have studied cross-sectionally. It resonates with a theme from an OECD working paper by [Tyros et al. \(2023\)](#) which argued that most workers in polluting (brown) occupations have skills compatible with greener jobs except those in highly automatable roles. Such analysis relies on comparing task requirements. The index can facilitate similar comparisons by quantifying how far apart occupations are in green task content.

Internationally, another approach worth noting is through job advertisements and big data. For example, an OECD paper by [Sato et al. \(2023\)](#) used online vacancies across multiple countries to identify “low-carbon” vs “high-carbon” jobs and the skills they require. They reported, consistent with others, that both low-carbon (green) and high-carbon (brown) jobs list higher skill requirements than generic jobs, and the gap between low-carbon and high-carbon is smaller than that between low-carbon and typical jobs([Sato et al., 2023](#)). In other words, high-carbon jobs are also often skilled (think petroleum engineers, etc.), which again implies workers in those jobs have a lot of human capital that can be redirected to green sectors with some retraining. They also noted limited geographic overlap of green and brown job postings, meaning regions tend to specialize in one or the other. This could pose regional transition challenges.

The GOJI measure doesn’t directly address skill level differences, but by linking to occupational attributes one can indirectly see these patterns (e.g., calculating average GOJI by occupational requirements level or by education level of workers in these occupations). Preliminary checks on German data indicate that occupations requiring a vocational or academic degree tend to have higher GOJIs than those not requiring formal training, which aligns with the notion that knowledge-intensive jobs have taken on more environmental tasks.

5.2 Labor Market Impacts of the Green Transition

A major question in both academic and policy circles is how moving towards a green economy affects jobs: Will it create or destroy jobs overall? Will it cause unemployment or just reallocation? Who are the winners and losers? Over the last 20 years, a number of studies have addressed these from macro and micro perspectives:

On the macro side, general equilibrium models and empirical macro studies have tried to estimate net employment effects of environmental policies like carbon pricing. [Hafstead and Williams \(2018\)](#) is a notable contribution using a two-sector search-and-matching model to simulate a carbon tax’s impact on employment. Their conclusion: a carbon tax causes substantial job shifts away from the polluting sector to the clean sector, but the *net* effect on overall unemployment is very small([Vona, 2023](#)). Their work, originally an NBER working paper, indicates that even reasonably large carbon taxes have been absorbed by economies without causing recessions or spikes in joblessness, partly because the revenue recycling and market adjustments help maintain employment.

However, the consensus of “small net effect” hides distributional impacts. Jobs lost in emissions-intensive industries might not be in the same locations or require the same skills as jobs created in green industries. This is where labor economists emphasize the importance of worker transition policies. [Bowen et al. \(2018\)](#) argued that the main issue is not the quantity of jobs but their matching with workers: reallocation frictions (like workers needing retraining or relocation) can cause temporary unemployment for those leaving brown industries.

Empirically, one can observe transition issues in specific sectors. The decline of coal mining is a prime example. Germany’s coal phase-out provides a case studied by [Haywood et al. \(2024\)](#), who use panel data of workers affected by lignite mine closures. An interesting finding was that many displaced coal workers found reemployment in other industries but often at lower wages or less job security([Haywood et al., 2024](#)). They stress that decarbonization policies should be accompanied by targeted support for affected workers, such as wage insurance schemes and retraining programs.

Another microcosm is the automobile sector’s shift to electric vehicles. This is ongoing,

but projections (e.g., by IW Consult and others in Germany) suggest overall auto industry employment might shrink, but new jobs in battery production and EV maintenance will grow. The workers may need different skills (EVs require more electrical/electronics knowledge, less traditional mechanics).

Recent cross-country evidence by (Bluedorn et al., 2023) looked at labor force survey data across many countries to see how greener economies differ. They constructed indices similar to GOJI using both industry emissions data and O*NET-style task data. Among their findings: countries with more flexible labor markets (easier hiring/firing, better training systems) saw a smoother reallocation of workers from polluting to greener jobs. In contrast, countries with rigid labor markets had more trouble, with workers getting stuck in declining brown industries or exiting the labor force. This suggests policy can mitigate pains of transition by fostering flexibility and mobility (e.g., portable pensions, housing mobility, etc.).

Additionally, Bluedorn et al. (2023) noted that greener jobs and polluting jobs each tend to concentrate among subsets of workers rather than being universal. This means many workers hold jobs that are largely neutral (low environmental impact), and relatively few work in extremely green or extremely polluting ones. That observation resonates with the GOJI distribution where more than half of occupations are classified as white and only the tails are heavily green/brown. This concentration implies that targeted measures could focus on those specific groups in occupations with brown tasks for transition assistance, but also in occupations in the white group or even in the green group, transitions to greener technologies and practice might be worthwhile to support.

Cecere and Mazzanti (2017) found that SMEs engaging in eco-innovations were more likely to increase their share of green employment over time. This finding is supported by Marin and Vona (2015), who demonstrated that firms investing in green R&D tended to experience employment growth, particularly when innovation targeted new markets. Similarly, Horbach and Janser (2016) found evidence that product-related environmental innovations can have a positive effect on employment, especially in firms with higher innovation capacity and environmental awareness.

Finally, a line of sociological and sustainability literature (often in journals like *Environmental Innovation and Societal Transitions*) discusses the concept of “just transition” – ensuring that the move to a green economy is fair for workers and communities. This is more qualitative but relevant. Studies here emphasize the role of governments and social partners in planning transitions (e.g., reskilling coal workers to renewable energy projects, as was somewhat attempted in Spain’s mining regions, etc.).

In summary, previous research provides several takeaways:

- *Measurement matters:* how I define green jobs can alter what I count, but task-based measures like GOJI are increasingly seen as useful for granularity.
- *Green jobs tend to be relatively skilled and not dramatically different from other jobs in skill requirements:* good for mobility, but also implies a need for skill development in certain areas.
- *Net employment effect of greening is small at macro level, but reallocation is significant:* policy should minimize friction, help workers in brown industries pivot to new roles.
- *Regional and demographic disparities exist:* targeted interventions may be needed to ensure certain groups (e.g., older workers, certain regions, women in STEM fields) are not left behind.
- *Continuous monitoring is needed:* as the transition accelerates (with more stringent climate policies ahead), tools like GOJI can help track how the job landscape is changing year by year.

This literature review underscores the relevance of the GOJI approach. By providing an ongoing, nuanced measure of occupational greenness, GOJI can feed into many of these discussions. For instance, if policymakers want to monitor whether workforce greenness is keeping up with policy targets, they can track the average GOJI or share of workers in green tasks over time. Researchers studying the labor impacts of specific policies (like a new environmental regulation) could use GOJI to classify which occupations are directly

affected (those with brown tasks related to the regulated pollutant) and then analyze outcomes.

In the next section, I conclude and reflect on how the GOJI can be applied and extended, connecting back to the themes identified here.

6 Conclusion

The Greenness-of-Jobs Index (GOJI) presented in this report offers a detailed lens through which to view the ongoing transformation of work in the face of environmental challenges. By focusing on tasks, GOJI moves beyond simplistic classifications of jobs as "green" or "not green" and acknowledges that most occupations lie on a continuum. This task-based approach has allowed me to quantify both the greening and browning of occupations over time, providing evidence that the transition to a sustainable economy is well underway in the labor market, albeit unevenly.

Several key insights emerge from this work:

- **Greenness is spreading, but incrementally.** Over the past decade, many occupations have incorporated more environmentally friendly tasks, raising their GOJI scores modestly. This indicates a broad but gradual greening of work. However, the majority of occupations still have GOJI values near zero, meaning the typical job is neither strongly green nor brown yet. The greening that has occurred is often in specialized tasks or roles at the margins of broader occupations.
- **Brown tasks remain prevalent in certain sectors.** A minority of occupations, particularly those in fossil fuel extraction, conventional energy, heavy manufacturing, and transport, carry a disproportionate share of brown tasks. These "brown" jobs are shrinking in some cases (e.g., coal mining) but persist, highlighting the continued dependence on carbon-intensive activities. GOJI allows me to pinpoint which occupations (and by extension, which workers) are likely to be most vulnerable in the transition, needing either upskilling or alternative employment as brown tasks phase out.

- **Net employment greening is positive but modest.** When weighting GOJI by employment (the share of all tasks in the economy that are green vs brown), I find an upward trend—suggesting that not only are jobs becoming greener, but greener jobs are becoming more common in terms of employment share. Still, the net measure is only slowly improving, reflecting that the pace of change may need to accelerate to meet ambitious climate targets. It also implies that a large workforce segment has yet to be touched by greening efforts, which could be both a challenge and an opportunity (untapped potential for greening many roles).
- **Task-based greenness aligns with skill and wage patterns.** Although my primary aim was measurement, the analysis suggests that occupations with higher GOJI tend to require higher skill levels and often offer higher wages, consistent with the literature that green jobs often involve knowledge-intensive activities. This means the green transition could, if unmanaged, exacerbate skill-biased employment trends. On the positive side, it means investing in education and training is doubly beneficial—raising human capital and enabling the green economy.
- **Policy can leverage GOJI findings.** For workforce development, GOJI can help identify which occupations should be targeted for green skill training programs (either because they are currently brown and need to change, or because they are emerging green occupations lacking sufficient skilled workers). For regional policy, mapping the average GOJI of jobs in different regions can indicate which local economies are more exposed to brown-job decline or which are gaining green jobs. For example, a region dominated by automotive manufacturing might have many occupations with moderate negative GOJI (related to combustion engine work); this region might benefit from proactive transition planning as the auto industry shifts to electric vehicles and battery tech (which would raise GOJI in those occupations if successful retraining is done).

The development of GOJI 2025 also provides some methodological lessons. I found that combining expert-driven keyword selection with iterative data-driven refinement is

an effective way to classify tasks. While my approach was applied to German data, the concept is transferable. Other countries with rich occupational information (such as Canada’s National Occupational Classification descriptions, or O*NET in the U.S., or ESCO in the EU context) could develop their own GOJI-like indices. Indeed, an interesting avenue for future research is to compare the greenness of jobs internationally: are German occupations greener or browner than their counterparts in other countries? Differences might reflect varying industry composition or policy strictness.

However, one must be cautious. GOJI measures the *intended* or *required* environmental content of jobs as described in official sources. It does not measure actual environmental performance (e.g., whether a company truly reduces emissions as a result of those tasks being present) nor the effort put by individual workers into those tasks. There may be instances of “window dressing” where job descriptions mention sustainability but in practice the role spends little time on it. Conversely, some workers may innovate greener practices that are not yet reflected in formal descriptions. Over time, job profiles will likely evolve to catch up with reality (or vice versa), but a lag can exist.

There is also the matter of tasks that might be environmentally relevant but are not captured by the keyword method due to subtle phrasing. While I have attempted to be comprehensive, language is nuanced and evolving. I anticipate updating the dictionaries periodically as new jargon and technologies emerge (for example, terms around hydrogen economy, carbon capture, etc., might become more common and need inclusion). The structure of GOJI allows for such updates.

Another limitation is that GOJI is inherently a measure at the occupation level, and thus cannot differentiate within an occupation. In reality, not every worker in a given occupation will perform the exact same tasks—there’s variation. For instance, an electrical engineer working for a coal power plant will have a different daily focus than an electrical engineer working for a wind turbine manufacturer, yet they might share the same occupation code. In such cases, GOJI provides an average. In the future, more granular data like job postings or employee surveys could be used to refine within-occupation variation, but that becomes complex. My stance is that occupational level is

a useful and tractable level of analysis for policy and forecasting.

The findings from GOJI 2025 highlight the importance of continuous workforce upskilling. Many brown tasks can be transformed into green tasks through training. For example, a truck mechanic today (brown-leaning) could be tomorrow's electric truck mechanic (green-leaning) if provided the right training in electric drivetrains. I see evidence of such transitions in the occupational descriptions themselves: new tasks get appended as technology changes. Facilitating workers to acquire those new tasks is critical. In this sense, GOJI could be used to monitor whether training curricula and vocational programs are updating their content to include these new green tasks. If an occupation's GOJI is rising but the training programs for that occupation are not yet teaching the corresponding green skills, a gap exists.

I also emphasize that a "just transition" requires not only creating green tasks/jobs but ensuring that the workers from brown jobs can move into them. Literature suggests skill overlap is often sufficient, but practical barriers like geographic mobility, age, or credentialling can still hinder transitions. Policies ranging from job search assistance, relocation support, apprenticeships in green tech for mid-career workers, to broader social safety nets will remain important.

In conclusion, the Greenness-of-Jobs Index provides a powerful tool for analyzing the labor market aspects of the environmental transition. It quantifies an otherwise abstract concept—how green is a job?—in a way that is comparable across occupations, regions, and time. As nations worldwide pursue decarbonization and sustainable growth, tools like GOJI can help track progress and inform policy adjustments. The evidence from GOJI 2025 is cautiously optimistic: economies can green themselves without massive job losses, but the transformation is slow and requires active support. The workforce of the future will likely look different, with new skills and tasks centered around sustainability. By measuring and understanding these shifts, I can better prepare for them, ensuring that the green economy works for everyone.

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A Appendix A: Classification of Tasks and Examples

A.1 Green Task Categories and Sample Keywords

The following are illustrative categories of green tasks and examples of keywords (in German, since the analysis is on German text) that signal those tasks in occupational descriptions:

- **Renewable Energy Tasks:** Tasks involving the production, installation, or maintenance of renewable energy technologies.
 - Keywords: *Solar* (e.g. Solaranlage, Solarzelle), *Windkraft/Windenergie* (wind power), *Photovoltaik* (photovoltaics), *Biogasanlage* (biogas plant), *Wasserkraft* (hydropower), *Geothermie* (geothermal).
 - Example Task: "*Montage und Wartung von Solarmodulen auf Dächern*" – "Installing and maintaining solar panels on roofs." (Green)
- **Energy Efficiency and Conservation:** Tasks aimed at reducing energy use or improving efficiency in systems and buildings.
 - Keywords: *Energieeinsparung* (energy saving), *Energieeffizienz* (energy efficiency), *Wärmedämmung* (thermal insulation), *energetische Sanierung* (energy retrofit), *LED-Beleuchtung* (LED lighting), *energieeffizient* (energy-efficient).
 - Example Task: "*Durchführen von Energieaudits in Gebäuden zur Steigerung der Energieeffizienz*" – "Conducting energy audits in buildings to improve energy efficiency." (Green)
- **Pollution Control and Environmental Protection:** Tasks focusing on monitoring, controlling, or preventing pollution and environmental harm.
 - Keywords: *Emission reduzieren* (reduce emissions), *Abgasreinigung* (exhaust gas cleaning), *Filteranlage* (filtration system), *Umweltschutz* (environmental protection), *Kläranlage* (sewage treatment plant), *Luftqualität messen* (measure air quality).

- Example Task: *”Überwachen von Emissionen und Einhalten von Grenzwerten in der Produktionsanlage”* – “Monitoring emissions and complying with limits in the production facility.” (Green)
- **Waste Management and Recycling:** Tasks related to managing waste streams, recycling materials, and promoting circular economy practices.
 - Keywords: *Recycling* (recycling), *Wiederverwertung* (reuse), *Abfallmanagement* (waste management), *Mülltrennung* (waste sorting), *Kompostierung* (composting), *Entsorgung nach Vorschrift* (disposal according to regulations).
 - Example Task: *”Planung eines Abfallmanagementkonzepts zur Abfallvermeidung und Recyclingsteigerung”* – “Planning a waste management concept to reduce waste and increase recycling.” (Green)
- **Environmental Planning and Management:** Tasks that involve incorporating environmental considerations into planning, management, and consultancy.
 - Keywords: *Umweltverträglichkeitsprüfung* (environmental impact assessment), *Nachhaltigkeitsbericht* (sustainability report), *Umweltmanagement* (environmental management), *ISO 14001* (an environmental management standard), *Naturschutz* (nature conservation).
 - Example Task: *”Beraten von Unternehmen bezüglich Nachhaltigkeitsstrategien und Umweltmanagementsystemen”* – “Advising companies on sustainability strategies and environmental management systems.” (Green)
- **Other Green Practices:** Tasks that promote sustainability in various domains not covered above (agriculture, transportation, etc.).
 - Keywords: *Ökologischer Landbau* (organic farming), *E-Mobilität* (e-mobility, electric vehicles), *Carsharing-Konzept* (car-sharing concept), *umweltfreundliche Materialien* (environmentally friendly materials), *Ressourcenschonung* (resource conservation).

- Example Task: *”Entwicklung von umweltfreundlichen Verpackungslösungen aus recyclebaren Materialien”* – “Developing eco-friendly packaging solutions from recyclable materials.” (Green)

These categories often overlap in practice (e.g., a task could involve both energy efficiency and pollution control). The keywords above are part of the lexicon used to flag green tasks in GOJI 2025. The actual implementation uses German terms as above, often with wildcards or regex patterns to capture variations (e.g., *”Energieeffizienz”* would also catch *”energieeffizient”*, etc.).

A.2 Brown Task Categories and Sample Keywords

Conversely, here are categories of brown tasks and examples of signals used to identify them:

- **Fossil Fuel Extraction and Processing:** Tasks involved in extracting coal, oil, gas, or processing them into fuel.
 - Keywords: *Kohlegewinnung* (coal extraction), *Bergbau* (mining), *Bohrung* (drilling), *Ölraffinerie* (oil refinery), *Fracking* (fracking).
 - Example Task: *”Bedienen von Maschinen zur Kohleförderung unter Tage”* – “Operating machines for underground coal mining.” (Brown)
- **Conventional Power Generation:** Tasks related to operating or maintaining fossil-fuel power plants and equipment.
 - Keywords: *Kraftwerk* (power plant, often contextually coal or gas), *Dampfturbine* (steam turbine), *Kesselwartung* (boiler maintenance), *Verbrennungsmotor* (combustion engine).
 - Example Task: *”Überwachung des Betriebs von Gaskraftwerksanlagen und Turbinen”* – “Monitoring the operation of gas power plant systems and turbines.” (Brown)

- **High Emissions Industrial Processes:** Tasks in industries known for very high energy use or emissions (steel, cement, chemicals).
 - Keywords: *Hochofenbetrieb* (blast furnace operation), *Stahlerzeugung* (steel production), *Zementofen* (cement kiln), *Petrochemische Anlage* (petrochemical plant).
 - Example Task: "*Steuern des Hochofens bei der Roheisenproduktion*" – "Controlling the blast furnace in pig iron production." (Brown)
- **Transportation and Combustion Engines:** Tasks involving the use of vehicles or machinery powered by fossil fuels.
 - Keywords: *Dieselmotor* (diesel engine), *Ottomotor* (gasoline engine), *Kraftstoffverbrauch* (fuel consumption), *Abgasausstoß* (exhaust emission).
 - Example Task: "*Warten von Dieselmotoren in Lastkraftwagen*" – "Servicing diesel engines in trucks." (Brown)
- **Use of Polluting Substances:** Tasks that involve handling or applying substances that are harmful to the environment.
 - Keywords: *Pestizide ausbringen* (apply pesticides), *Industriechemikalien* (industrial chemicals), *Lösemittel einsetzen* (use solvents), *Gefahrstoff* (hazardous material).
 - Example Task: "*Ausbringen von Düngemitteln und Pestiziden auf landwirtschaftlichen Flächen*" – "Applying fertilizers and pesticides on agricultural land." (Brown)
- **Environmental Damage or Non-compliance:** Tasks that explicitly mention environmentally harmful outcomes or lack of mitigation.
 - Keywords: *Umweltbelastung* (environmental burden), *ohne Abgasfilter* (without exhaust filter), *Abfälle verbrennen* (burn waste), *Gewässerverschmutzung* (water pollution) – note these might appear rarely in descriptions, but if they do, it's a clear brown flag.

- Example Task: (Hypothetical, as overtly negative phrasing is rare in official descriptions) *”Verbrennen von Industrieabfällen im Freien”* – “Burning industrial waste in the open.” (Brown)

In occupational descriptions, brown tasks are often described in neutral terms (e.g., “operate diesel generator”), so the context of what’s implied is important. I used keywords that directly point to fossil fuel usage or pollutant-related processes. As mentioned, I avoid misclassifying tasks that are about mitigating these issues as brown by carefully crafting patterns (for example, a task containing *”Abgasemissionen reduzieren”* (reduce exhaust emissions) is green, whereas *”Abgasemissionen messen”* (measure exhaust emissions) might be neutral or brown depending on if it’s for control or just part of operation).

A.3 White Task Examples

For completeness, I list a few examples of common white tasks (these would not trigger either green or brown classification):

- **Administrative Tasks:** e.g. *”Verwaltung der Projektdokumentation und Führen von Protokollen”* – “Managing project documentation and keeping minutes.” (Neutral)
- **Generic Technical Tasks:** e.g. *”Durchführung von Qualitätskontrollen am Produkt”* – “Conducting quality checks on the product.” (Neutral, unless the quality check specifically relates to an environmental parameter).
- **Customer Service:** e.g. *”Beratung von Kunden zu Produktfunktionen und Serviceleistungen”* – “Advising customers on product features and services.” (Neutral, assuming product isn’t specifically an environmental product).
- **Routine Operations:** e.g. *”Bedienen der Produktionsanlage gemäß Arbeitsanweisung”* – “Operating the production machinery according to work instructions.” (Neutral if no environmental aspect in instructions).

- **General Maintenance:** e.g. *”Durchführen von Reparaturen und Instandhaltungsarbeiten an Maschinen”* – “Performing repairs and maintenance on machines.” (Neutral unless machine context is specifically an enviro-tech or a pollutant source highlighted in description).

White tasks make up the bulk of most occupations’ task lists. They represent the core duties that, while essential for the job, don’t have a direct environmental angle.

A.4 Occupation Vignette

To illustrate how tasks combine in an occupation, Table 1 provides a fictional but representative example of an occupation with its tasks categorized:

Table 1: Example Occupation – *Industrial Machinery Mechanic* (Fictional Tasks)

Task Description (translated from German)	Category	Flagged as
Inspect and maintain factory machinery and production lines.	Neutral	–
Repair and replace faulty mechanical parts and motors.	Neutral	–
Optimize machine settings to reduce energy consumption during operation.	Green	(Green task)
Lubricate and cool machine components using standard oils and coolants.	Brown (due to oils)	(Brown task)
Implement a schedule for preventive maintenance to minimize downtime and energy waste.	Green	(Green task)
Document maintenance activities and report any irregularities.	Neutral	–

In Table 1, the occupation has 6 listed tasks. Using GOJI classification: - Green tasks: 2 out of 6 (tasks about optimizing for energy efficiency, and preventive maintenance with energy considerations). - Brown tasks: 1 out of 6 (task involving use of standard oils/coolants, which I assume is flagged because of resource use/potential pollution). - Neutral: 3 out of 6. $GOJI^+ = 2/6 \approx 0.333$, $GOJI^- = 1/6 \approx 0.167$, net $GOJI \approx 0.167$. This would categorize as a light green occupation.

Without the green optimization tasks, this occupation would be neutral or slightly

brown (traditional maintenance). The example shows how adding just a couple of green-oriented tasks (energy optimization, etc.) can shift an occupation into green territory. In reality, such shifts often happen as industries adopt new standards (for instance, many industrial companies now include “optimize energy usage” in maintenance job roles, which wouldn’t have been there 20 years ago).

B Appendix B: Coding Instructions and Algorithm Details

This appendix provides a more technical outline of how the GOJI coding was implemented, which may be useful for replication or extension:

B.1 Preprocessing and Pattern Matching

1. **Data Ingestion:** Occupational descriptions were obtained in text form for each year. I structured them into a dataset with columns: `occupation_code`, `year`, `task_text`.
2. **Text Cleaning:** For each `task_text`, I applied:
 - Lowercasing.
 - Removal of punctuation (except those like hyphens if relevant for terms).
 - Replacement of German umlauts with their two-letter equivalents (ä-ıae, ö-ıoe, ü-ıue, ß-ıss) *or* inclusion of them in regex patterns using Unicode classes.
 - Tokenization for some approaches (though regex allowed phrase matching without full tokenization).
3. **Regex patterns:** I compiled two lists of regex patterns (one for green, one for brown). Patterns ranged from single words to phrases. For example:
 - Green regex sample: `\benergiespar` would match “energiesparende” (energy-saving) etc. Another: `\bnachhaltig` for words starting with `nachhaltig` (sus-

tainable).

- **Brown regex sample:** `\bkohle` for coal, `verbrennungsmotor` for combustion engine.
 - **Context-sensitive patterns:** I used negative lookahead or lookbehind where needed. e.g. pattern for emissions as brown might be `Emission(?!.*reduz)` which matches "Emission" unless the word "reduz(ieren)" (reduce) appears later in the same task string, to avoid tagging "reduce emissions" as brown.
4. **Matching procedure:** For each task text, I applied all green patterns. If any pattern matched, I marked the task as green and did not test it against brown patterns (to avoid double classification). If none green matched, I then applied brown patterns. If a brown pattern matched, marked as brown. If neither matched, it remained neutral.
5. **Edge cases:** A few tasks contained multiple sentences or were long; I considered a match anywhere in the task text as enough to classify the whole task. There were instances where a task could conceptually be both (e.g., "monitor emissions from diesel generators to ensure legal compliance" – monitoring emissions is neutral or green if ensuring compliance, but "diesel generator" is brown context; the rules likely classify that as green because it's about compliance and monitoring, an environmental management aspect).

B.2 Quality Checks and Iteration

I performed the following checks:

- **Manual review:** took random samples of tasks classified as green, brown, and neutral to verify they made sense. Adjusted patterns if I found systematic errors (e.g., a green term appearing in a non-environment context).
- **Count validation:** Ensured that for each year, $G_{o,t} + B_{o,t} \leq T_{o,t}$ and examined cases of equality (if any occupation had every task flagged as either green or brown).

Usually, some tasks remain neutral, but a very green-focused occupation could indeed have all tasks green (e.g., an environmental protection officer might have 100

- **Overlap analysis:** Checked if any occupation had both $G_{o,t}$ and $B_{o,t}$ relatively high (≥ 0 in both). There were some, which were often technical roles that do a mix (e.g., automotive engineers who work on conventional engines (brown) and also on emissions reduction tech (green)). These cases are interesting as they highlight transitional roles. None had extremely high on both simultaneously.
- **Trend consistency:** I expect that from 2011 to 2022, most occupations would stay same or increase in GOJI (because new green tasks added or no change; it's less common to add new brown tasks over time in descriptions, as industries rarely introduce new polluting processes without also mentioning controls). I found this generally true – occupations either held steady or went up in net GOJI. A few fluctuations occurred (some small drops, possibly due to description changes or reclassification of tasks) but no systematic greening reversal.

B.3 Comparison of GOJI Versions (2018 vs. 2025) Methodologically

For reference, Table 2 summarizes key differences among GOJI 2018, the Beta, and GOJI 2025 methodologies:

Table 2: Comparison of GOJI Methodologies

Aspect	GOJI (Original)	2018 GOJI Beta (circa 2021)	GOJI 2025 (Cur- rent)
Task categories	Green tasks only; others implicitly non-green	Green and Brown tasks (brown defined narrowly)	Green, Brown, and White tasks explicitly categorized
Index components	<i>GOJI</i> : share of green tasks	<i>GOJI</i> ⁺ and <i>GOJI</i> ⁻ shares, plus net index	<i>GOJI</i> ⁺ , <i>GOJI</i> ⁻ , net <i>GOJI</i> (Green minus Brown share)
Keywords for Green	~300 keywords (emphasis on renewables, efficiency)	largely same	Expanded to 399 keywords (added e-mobility, etc.)
Keywords for Brown	Not used	~100 keywords (strict: fossil fuels, etc.)	Expanded to 191 keywords (broader, incl. engines, chemicals)
Data years	2006, 2011–2016 (selected)	2011–2020 (annual, internal)	2011–2022 (annual)
Occupational detail	KldB 2010 5-digit (some aggregated from 8-digit descriptions)	KldB 2010 5-digit	KldB 2010 5-digit (consistent)
Text-mining method	Lexicometric content analysis (frequency-based, initial word list)	Regex match (improved word list), but cautious on brown	Regex match (refined lists, context rules)
Validation	Internal consistency checks, face validity of top green jobs	Preliminary validation, some expert review	Extensive validation (manual checks, iterative refinement)
Output	Single index per occ (0–1)	Two indices + net (can be negative)	Two indices + net (range -1 to +1)
Classification use	Descriptive stats, correlation with employment growth	Used in research on decomposition of greening (within vs between occ)	Used for descriptive analytics, identifying categories (dark green/brown, etc.)

C Appendix C: Additional Descriptives

C.1 Distribution of GOJI (Net) in 2012 and 2022

Figure 1 and 2 illustrate the distribution of the net GOJI across occupations for the years 2012 and 2024. Occupations are binned into the five categories defined in the text.

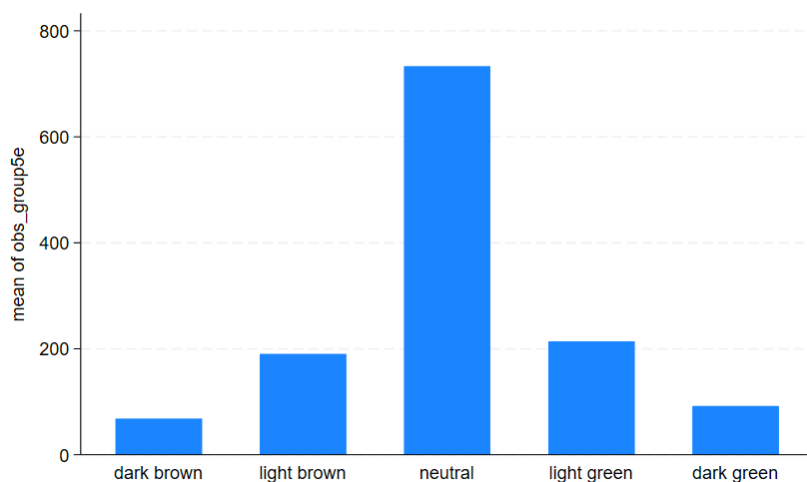


Figure 1: Histogram of Occupational GOJI Values in 2012. Dark brown occupations ($\text{GOJI} < -0.1$) and dark green occupations ($\text{GOJI} > 0.1$) constitute the tails. (Source: BERUFENET)

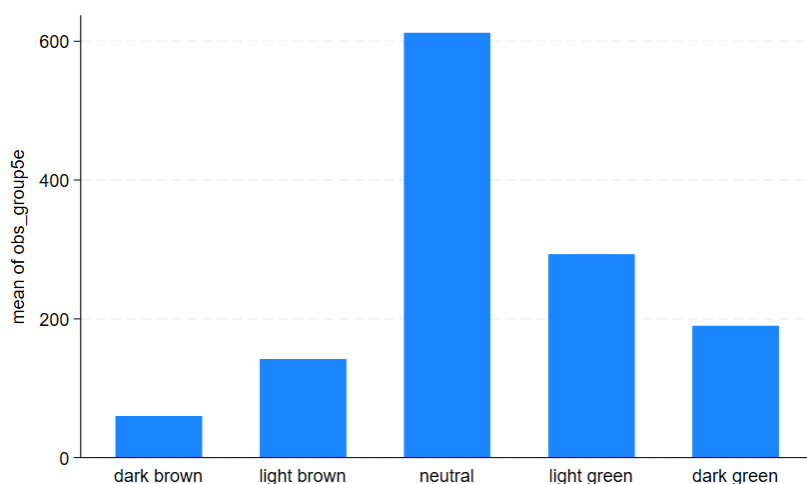


Figure 2: Histogram of Occupational GOJI Values in 2024. Dark brown occupations ($\text{GOJI} < -0.1$) and dark green occupations ($\text{GOJI} > 0.1$) constitute the tails. Compared to 2012, several occupations shift slightly to the right by 2024, indicating overall greening. (Source: BERUFENET)

D Appendix D: Availability of GOJI in the IAB Occupational Panel

The GOJI is integrated in the publicly accessible IAB Occupational Panel, which is released under the Creative Commons Attribution-ShareAlike 4.0 International (CC BY-SA 4.0) license.¹ The most recent version (IAB Occupational Panel 2012–2022, referred to as `iab-occpan1222v2`) includes GOJI 2.0 (beta) for the period 2012–2022, at the 3+5 level of the KldB 2010 classification (i.e.:. The documentation of the prior Panel version is provided by Grienberger, Janser–Lehmer (2022), while changes in the 2022 update—including the incorporation of GOJI and related metadata—are listed in the Panel’s change log (IAB 2025).² In the upcoming version of the IAB Occupational Panel, covering the years 2012 to 2025, the GOJI will also be included using its most recent version.

¹The IAB Occupational Panel can be downloaded here: <https://iab.de/en/daten/iab-occupational-panel/>.

²https://doku.iab.de/arbeitsmarktdaten/ChangeLog_OccPan1222v2_EN_20250311.txt.